

Furnish3D — Project Report

Interactive 3D Furniture Store

Field	Details
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Course	LLM Driven Web Mastery
Company	Glaxit
Project	Furnish3D – Interactive 3D Furniture Store
Submission date	August 27, 2026

1. Quick links (for instructor)

Resource	URL
Live website	https://frontend-ten-iota-75.vercel.app
Live API	https://furnish3d-api-production.up.railway.app
API health check	https://furnish3d-api-production.up.railway.app/api/health
GitHub repository	https://github.com/muhammadibrahim14375-bot/Furnish3D

Demo accounts (password for all: password123)

Email	Role	What to try
customer@furnish3d.com	Customer	Browse, 3D viewer, cart, orders, reviews
mod@furnish3d.com	Moderator	Approve reviews, edit products
admin@furnish3d.com	Admin	Full product/user/order management

2. Project overview

Furnish3D is a full-stack furniture e-commerce web application. Unlike traditional stores that only show static photos, customers can **rotate and inspect furniture in interactive 3D** before purchasing.

The platform also includes secure authentication, role-based access (customer / moderator / admin), shopping cart, orders, product reviews, search/filtering, and a responsive dark/light theme UI.

3. Problem statement

Online furniture shopping usually depends on 2D images and text descriptions. Size, shape, and appearance from different angles are hard to judge that way. Furnish3D improves this by combining standard e-commerce features with an interactive 3D product viewer.

4. Objectives achieved

- ☒ Modern furniture e-commerce platform
- ☒ Interactive, rotatable 3D product previews
- ☒ Secure authentication with JWT and bcrypt
- ☒ Role-based access control (customer, moderator, admin)
- ☒ Product browsing, search, cart, orders, and reviews
- ☒ Moderator and administrator management panels
- ☒ Responsive UI with dark/light themes
- ☒ Frontend and backend deployed online

5. Technology stack

Layer	Technologies
Frontend	React, TypeScript, Vite, React Router, Three.js, React Three Fiber, Drei
Backend	Node.js, Express.js (MVC architecture)
Auth	JWT, bcrypt, RBAC middleware
Database	File-based store for live demo + PostgreSQL/Supabase schema (backend/sql/schema.sql)
Deployment	Frontend on Vercel , Backend on Railway

6. Main features

Customer

- Register / login
- Browse and search products by category
- Product detail page with **interactive 3D viewer** (drag to rotate, scroll to zoom)
- Add to cart, checkout, view order history
- Submit product reviews
- Toggle dark / light theme (preference saved)

Moderator

- Approve or reject pending reviews
- Edit product listings
- No user-management privileges

Admin

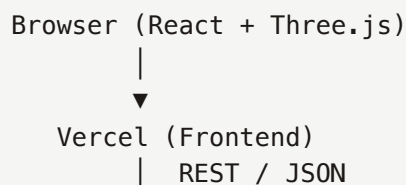
- Full product CRUD
- User list, role assignment, user deletion
- Order list and status updates
- Complete administrative control

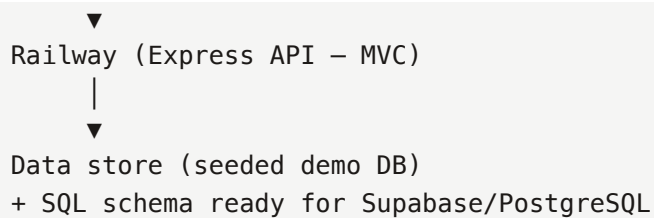
7. 3D product visualization

The main differentiator is the 3D viewer on each product page, built with **Three.js**, **React Three Fiber**, and **Drei**.

- Users can orbit and zoom around the furniture model
- Category-based procedural 3D models are used for reliable performance in the demo
- Product records also support GLB/GLTF `modelUrl` for real mesh assets later

8. System architecture





Backend modules (MVC)

- **Auth** — register, login, /me
- **Users** — admin user/role management
- **Products & categories** — browse, CRUD
- **Cart** — add / update / remove
- **Orders** — checkout, history, status
- **Reviews** — create, moderate, delete

9. How to run locally

```
# Backend
cd backend
npm install
npm run seed
npm run dev
# → http://localhost:5001

# Frontend
cd frontend
npm install
npm run dev
# → http://localhost:5173
```

10. Deployment summary

Part	Platform	Live URL
Frontend	Vercel	https://frontend-ten-iota-75.vercel.app
Backend	Railway	https://furnish3d-api-production.up.railway.app
Source code	GitHub	https://github.com/muhammadibrahim14375-bot/Furnish3D

11. Suggested instructor walkthrough

1. Open the **live website** link.
 2. Browse **Shop** and open any product → try the **3D viewer**.
 3. Log in as `customer@furnish3d.com / password123` → add to cart → place an order.
 4. Log in as `mod@furnish3d.com` → open **Moderate** → review moderation / product edit.
 5. Log in as `admin@furnish3d.com` → open **Admin** → products, users, orders.
 6. Review source code on **GitHub**.
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12. Conclusion

Furnish3D delivers a complete full-stack e-commerce experience with a practical 3D viewing feature where it adds real value for furniture shoppers. It demonstrates skills in React, TypeScript, REST APIs, Express MVC, authentication, RBAC, and Three.js, and is deployed for live evaluation.

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